

Geek Home (Home Automation)

Outcome:

To develop a home automation system using ESP32-C6 that allows control of appliances (Light, Fan, Motor, TV) via capacitive touch sensor and web interface.

Components:

S. No	Component	Quantity
1.	ESP32-C6 Dev Board	1
2.	5v Relay Module	1
3.	Jumper Wires	As needed
4.	USB Type-C Cable	1
5.	Computer with Arduino IDE	1
6.	Touch Sensor	1

Concept:

This system uses the ESP32-C6 microcontroller's built-in Wi-Fi capabilities and external input device touch sensor to control home appliances through:

- **Touch Sensor:** A capacitive input toggles the light relay.
- **Web Interface:** A local HTML-based control panel served over Wi-Fi allows toggling of each device.

Relays are connected to the ESP32-C6 GPIOs to switch high-voltage appliances.

Connections:

1. Circuit Assembly

- Connect the touch sensor to GPIO10.
- Connect each relay channel to GPIO7 respectively.
- Ensure relays share a common GND with the ESP32-C6.
- Connect the relay VCC and GND to Battery positive and negative wire

2. Upload the Code

- Use Arduino IDE or ESP-IDF to flash the given code to the ESP32-C6.
- Ensure required libraries are installed: WiFi.

3. Web Control Setup

- Connect ESP32-C6 to your Wi-Fi network using SSID and password.
- Monitor the Serial Monitor to get the IP address.
- Open this IP in a browser to access the web-based control panel.

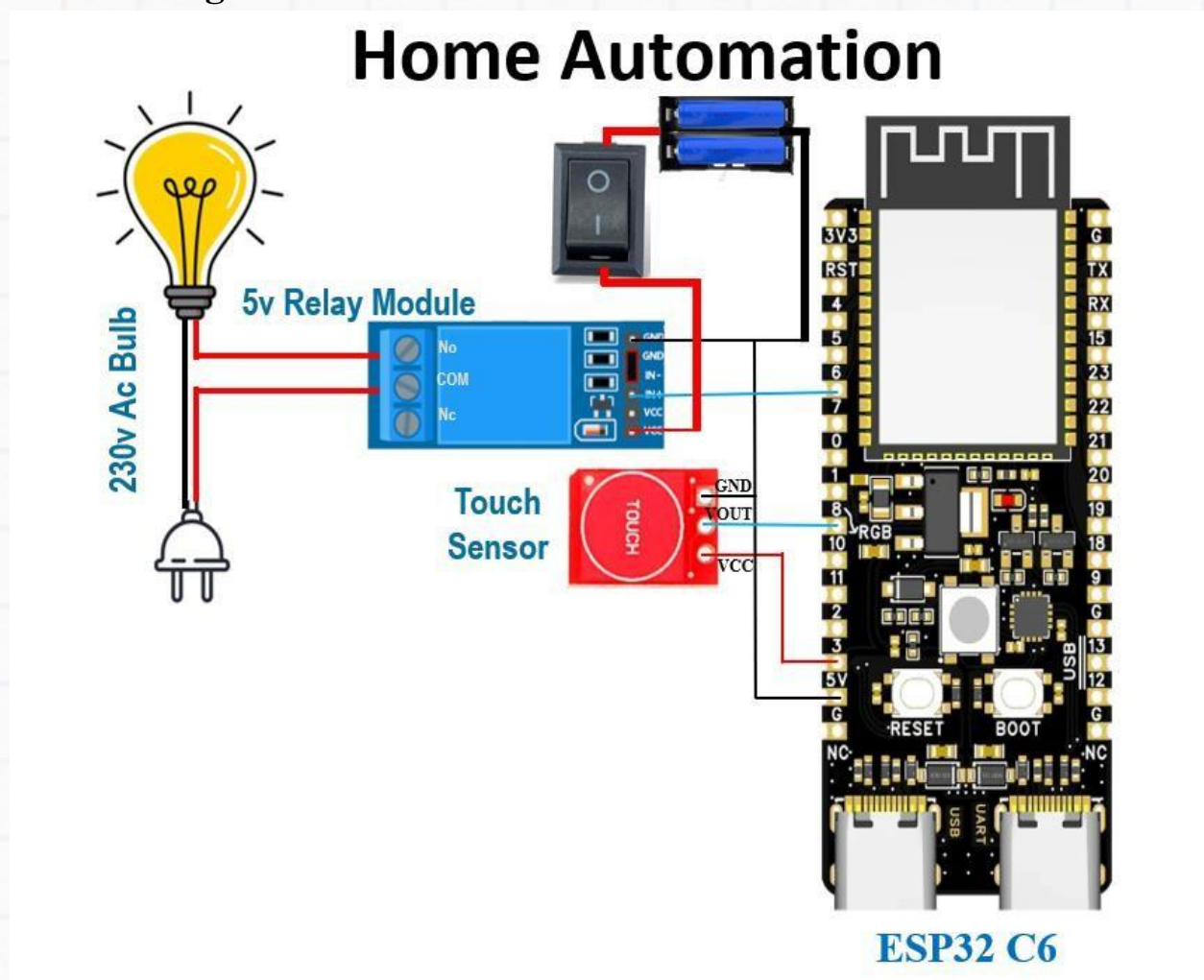
4. Touch Control

- Tapping the capacitive touch pin toggles, the Light relay only.
- A 500ms debounce avoids accidental triggering.

Prior learning:

- HTML - [Reference link](#)
- CSS - [Reference link](#)
- JavaScript - [Reference link](#)

Circuit Diagram:



Software Code :

```
#include <WiFi.h>
#include <WebServer.h>

const char* ssid = "1234";           // Replace with your Wi-Fi SSID
const char* password = "123456789m"; // Replace with your Wi-Fi password

#define RELAY_PIN 7
#define TOUCH_PIN 10

WebServer server(80);

bool relayState = false; // false = OFF, true = ON
bool lastTouchState = LOW;

void handleRoot() {
    String html = "<!DOCTYPE html><html><head><meta charset='UTF-8'>";
    html += "<meta name='viewport' content='width=device-width, initial-";
    html += "scale=1.0'>";
}
```

```

html += "<title>BUILD CLUB</title>";
html += "<style>body{font-family:sans-serif;text-align:center;padding:20px;}";
html += ".toggle{width:60px;height:30px;position:relative;display:inline-block}";
html += ".toggle input{display:none}";
html +=
".slider{position:absolute;cursor:pointer;top:0;left:0;right:0;bottom:0;";
html += "background-color:#ccc;transition:.4s;border-radius:30px;}";
html +=
".slider:before{position:absolute;content:'';height:22px;width:22px;";
html += "left:4px;bottom:4px;background-color:white;transition:.4s;border-radius:50%;}";
html += "input:checked+.slider{background-color:#2196F3;}";
html += "input:checked+.slider:before{transform:translateX(30px);}</style>";
html += "<h1>BUILD CLUB</h1>";
html += "<h1>Home Automation </h1>";
html += "<label class='toggle'><input type='checkbox'";
html += "onclick='toggleRelay(this)''";
html += "if (relayState) html += \" checked\"";
html += "><span class='slider'></span></label>";
html += "<p>Status: <span id='status'>";
html += (relayState ? "ON" : "OFF");
html += "</span></p>";
html += "<script>function toggleRelay(el){";
html += "var xhr=new XMLHttpRequest();";
html += "xhr.open('GET', el.checked ? '/on' : '/off', true);";
html += "xhr.send();}";
html +=
"setInterval(function(){fetch('/status').then(r=>r.text()).then(t=>{";
html += "document.getElementById('status').innerText=t;";
html += "document.querySelector('input').checked=(t==='ON');";
html += "});},1000);</script>";
html += "</body></html>";
server.send(200, "text/html", html);
}

void handleOn() {
    relayState = true;
    digitalWrite(RELAY_PIN, HIGH);
    server.send(200, "text/plain", "OK");
}

void handleOff() {
    relayState = false;
    digitalWrite(RELAY_PIN, LOW);
    server.send(200, "text/plain", "OK");
}

```



```

void handleStatus() {
    server.send(200, "text/plain", relayState ? "ON" : "OFF");
}
void setup() {
    Serial.begin(115200);
    pinMode(RELAY_PIN, OUTPUT);
    pinMode(TOUCH_PIN, INPUT);
    digitalWrite(RELAY_PIN, LOW); // Initially off

    WiFi.begin(ssid, password);
    Serial.print("Connecting to WiFi");
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
    }
    Serial.println("\nWiFi Connected!");
    Serial.print("IP Address: ");
    Serial.println(WiFi.localIP());

    server.on("/", handleRoot);
    server.on("/on", handleOn);
    server.on("/off", handleOff);
    server.on("/status", handleStatus);
    server.begin();
    Serial.println("HTTP server started");
}

void loop() {
    server.handleClient();

    bool touch = digitalRead(TOUCH_PIN);
    if (touch != lastTouchState && touch == HIGH) {
        relayState = !relayState;
        digitalWrite(RELAY_PIN, relayState ? HIGH : LOW);
        Serial.println(relayState ? "Relay ON by Touch" : "Relay OFF by Touch");
        delay(300); // debounce
    }
    lastTouchState = touch;
}

```

Post Workshop Tasks (Club Activities):

1. Build a Voice Controlled LED bulb that switches ON & OFF based on user input from phone.
2. Build a PIR based automated light that switches ON when someone enters a room & switches OFF when there is no movement.